# *Principia Mathematica* to ∗56

## A Commentary

with a Technical Retrospective on the Logic Theorist,
the first program to prove its theorems by machine

A hybrid reader's guide and critical commentary

September 23, 2026

# Preface

This volume gathers, in one place, a commentary on the abridged *Principia Mathematica to* $*56$— the logical core of Whitehead and Russell's system— together with a technical retrospective on the Logic Theorist, the 1956 program of Newell, Shaw, and Simon that was the first to prove theorems of *Principia* by machine. The pairing is deliberate. The commentary reads *Principia* forwards, as a human might; the retrospective shows the same propositional calculus read by a searching machine. Between them they look at one book from two sides—as a monument of thought and as a problem environment for mechanized reason.

The commentary is a hybrid of reader's guide and scholarly commentary. Each chapter opens with motivation—what the authors are trying to secure, and why it comes where it does—and continues with commentary on the pivotal propositions, restated in modern notation, with attention to proof strategy, historical context, and the connections to later logic. The propositional chapter ($*1$–$*5$) is truth-functional, and every proposition cited in it has been verified mechanically by the reconstruction described in the opening chapter; the later chapters, which involve quantifiers, types, and classes, rest instead on the primary text and the secondary literature, as their closing notes record. Star-number citations follow the standard second-edition numbering.

The five chapters may be read independently, but they tell a single story: from five propositional axioms and one rule of inference, through quantification and the theory of types, the theory of descriptions, and the extensional logic of classes and relations, to the definition of the first cardinal numbers and the threshold of arithmetic—the point at which, famously, *Principia* stands ready to prove that $1 + 1 = 2$.

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# Chapter 1

# The Logic Theorist: A Technical Retrospective

*The* Logic Theorist *(LT), built by Allen Newell, J. Clifford Shaw, and Herbert A. Simon at the RAND Corporation and the Carnegie Institute of Technology in 1955–56, is widely regarded as the first computer program engineered to perform non-numerical, human-like reasoning—the "first artificial-intelligence program." Its task domain was the propositional calculus of Whitehead and Russell's* Principia Mathematica *(PM): it proved* 38 *of the first* 52 *theorems of Chapter 2 and found a proof of theorem* ∗2.85 *shorter than the one printed in PM. This paper reconstructs how LT worked—its object language and list representation, the five PM axioms and three rules of inference it inherited, the four proof-finding* methods *and the* executive *that scheduled them, and the heuristic "similarity test" that kept its search tractable—and situates the machine against the logical monument it adopted as a problem environment. We argue that LT took over PM's formal system almost unchanged while inverting its intellectual purpose: where PM sought to* ground *mathematics in logic through complete, justificatory proof, LT treated proof* discovery *as an empirical, heuristic, and psychological process. A worked example—the derivation of* ∗2.85 *found by the reconstruction—makes the mechanism concrete.*

## 1.1 Introduction

In the winter of 1955–56 a collaboration between a RAND systems programmer, J. Clifford Shaw, and two researchers who would each become central figures in cognitive science and artificial intelligence, Allen Newell and Herbert Simon, produced a program whose ambition was unusual for its moment. Most computing of the period was numerical; the Logic Theorist was written to do something that had previously belonged to mathematicians—to *prove theorems*, and to do so not by an exhaustive decision procedure but in a manner its authors explicitly intended to resemble human problem solving. Simon later described the intellectual event with characteristic immodesty, telling a class early in 1956 that over the holiday he and Newell had "invented a thinking machine." Before a line of it ran on a computer, the program was hand-simulated: Newell and Simon wrote its steps onto index cards and had family members and graduate students each execute one card, so that the group as a whole enacted the program.

The choice of *Principia Mathematica* as a proving ground was, by Simon's own account, partly accidental—a copy sat on his shelf—and by then the ramified system of PM was regarded as outmoded for foundational work. But the propositional fragment of PM was ideal for the purpose at hand: a small, completely formal task environment with a fixed vocabulary, a handful of axioms, sharply defined rules, and a graded sequence of theorems of increasing difficulty. LT was implemented in the Information Processing Language (IPL), a list-processing language

Newell and Shaw devised for it and which is a direct ancestor of LISP, and it ran on the JOHNNIAC, RAND's copy of the Institute for Advanced Study machine. It was demonstrated at the 1956 Dartmouth Summer Research Project on Artificial Intelligence, the meeting that gave the field its name.

It is easy, from the far side of seventy years of progress, to underestimate how strange the undertaking was. Computers of the mid-1950s were understood as engines for arithmetic; the proposal that one might be programmed to *discover* a proof, and moreover to do so in a way modeled on human thought rather than on a mathematician's idealized decision procedure, ran against the grain of the machine's conception. LT was a wager that reasoning is a species of information processing, and that a symbolic process running on a computer could exhibit it.

This paper is a technical reconstruction. Section 1.2 sets out the propositional calculus of PM that LT inherited. Section 1.3 describes how LT represented logical expressions as list structures. Section 1.4 states the three rules of inference LT could apply, and Section 1.5 the four *methods*—substitution, detachment, and forward and backward chaining—by which it reduced a goal to subproblems. Section 1.7 describes the executive routine and the heuristics, above all the similarity test, that made the search feasible. Section 1.8 works through the machine's proof of $*2.85$. Sections 1.9 and 1.10 then step back to consider LT's relationship to PM and its place in the subsequent history of ideas.

## 1.2 The propositional calculus of *Principia*

*Principia Mathematica* (1910–13) is a defense of *logicism*, the thesis that mathematics is reducible to logic. Its opening numbers, $*1$ through $*5$, develop the propositional calculus that serves as the base of the whole edifice, and it is exactly this fragment that LT worked in.

**Language.** The primitive connectives of PM are negation $\neg$ and disjunction $\vee$. Implication is not primitive but *defined*,

$$A \supset B \quad \stackrel{\text{def}}{=} \quad \neg A \vee B, \tag{$*1.01$}$$

and conjunction and equivalence are defined in turn from these. A well-formed formula is a propositional variable ($p, q, r, \dots$) or is built from well-formed formulas by $\neg$ and $\vee$. A historical peculiarity, inherited from Russell's earlier work, is that although $\neg$ and $\vee$ are the only primitives, the axioms themselves are stated using the *defined* sign $\supset$.

**Axioms.** PM takes five "primitive propositions" for the propositional calculus:

$$\begin{array}{lll} *1.2 & \text{(Taut)} & (p \vee p) \supset p \\ *1.3 & \text{(Add)} & q \supset (p \vee q) \\ *1.4 & \text{(Perm)} & (p \vee q) \supset (q \vee p) \\ *1.5 & \text{(Assoc)} & \big(p \vee (q \vee r)\big) \supset \big(q \vee (p \vee r)\big) \\ *1.6 & \text{(Sum)} & (q \supset r) \supset \big((p \vee q) \supset (p \vee r)\big) \end{array}$$

**Rules.** Two rules of inference generate theorems from the axioms. The first is *modus ponens*, or detachment ($*1.1$): from $A$ and $A \supset B$, infer $B$. The second is *substitution*: from a theorem $A$ one may infer any formula obtained by replacing a propositional variable throughout $A$ by an arbitrary formula. A third operation, licensed by the definitions, is the *replacement* of any expression by a definitional equivalent—most importantly the interchange of $A \supset B$ with $\neg A \vee B$.

**Metatheory.** Although PM predates the modern semantic account of propositional logic, that account was supplied shortly afterward. Post (1921) proved the system consistent and complete in the truth-functional sense: a formula is derivable if and only if it is a tautology. Bernays (1926) showed that the axiom set is not independent—∗1.5 (Assoc) is derivable from the remaining four—so PM's propositional basis could be reduced to four axioms. These results matter for placing LT: the target theory is *complete*, so every truth-functional validity has a proof; but, as we shall see, LT made no use of completeness and offered no corresponding guarantee. It sought *a* proof by plausible means, not *the* proof guaranteed to exist.

## 1.3 Representation of expressions

LT's most consequential engineering decision was to represent logical expressions not as character strings but as *list structures*. Each formula was stored as a hierarchy of *elements*: the main connective was the top element, and its operands were sub-elements, themselves expressions with their own main connectives, recursively down to the variables at the leaves (Figure 1.1a). An implication $A \supset B$, for instance, is an element whose main connective is $\supset$, whose left sub-element is the antecedent $A$, and whose right sub-element is the consequent $B$.

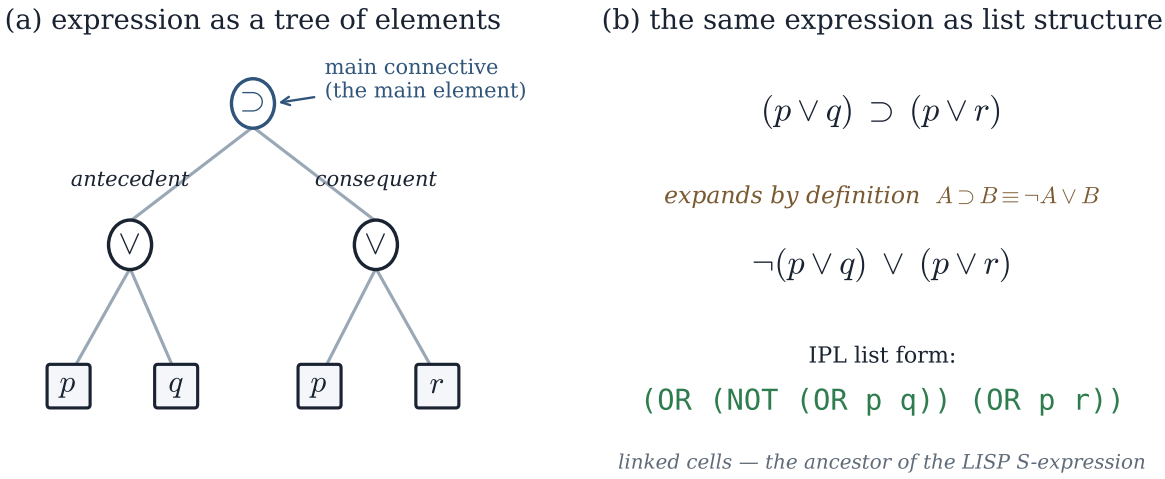



**Figure 1.1:** An expression as (a) a tree of elements and sub-elements and (b) the corresponding nested list. LT stored formulas as linked list cells of exactly this kind; the notation is the direct ancestor of the LISP S-expression.

This representation, expressed in IPL as chains of linked cells, made the structural operations of logic—finding the main connective, extracting a subexpression, substituting for a variable, testing two expressions for identity—into simple list manipulations rather than fragile string surgery (Figure 1.1b). List processing was invented for precisely this need, and the idea propagated: McCarthy's LISP a few years later generalized the same conception of symbolic computation over list structure. Alongside each expression LT also kept a cheap *description*—a small tuple of features such as the number of distinct variables and the depth of nesting—that its heuristics would consult without having to inspect the full expression. We return to these descriptions in Section 1.7.

## 1.4 The three rules as operations

At the level of legal inference LT could do exactly three things, corresponding to PM's rules:

1. **Substitution.** Replace every occurrence of a variable in a known theorem by a chosen expression, uniformly. This specializes a schematic theorem to the case at hand.
2. **Replacement.** Exchange a subexpression for its definitional equivalent, chiefly $A \supset B \leftrightarrow \neg A \vee B$. This lets the same formula be viewed either as an implication or as a disjunction, according to which is useful.
3. **Detachment.** From $A$ and $A \supset B$, infer $B$ (modus ponens).

These three operations are *sound* but say nothing about *search*: they tell us which steps are legal, not which to take. The intelligence of LT lay one level up, in the methods that decided how to bring these operations to bear on a goal.

## 1.5 The four methods

LT worked *backward* from the theorem to be proved. Given a goal, its executive tried up to four methods in a fixed order; each either closed the goal outright or replaced it with a simpler subproblem (Figure 1.2).

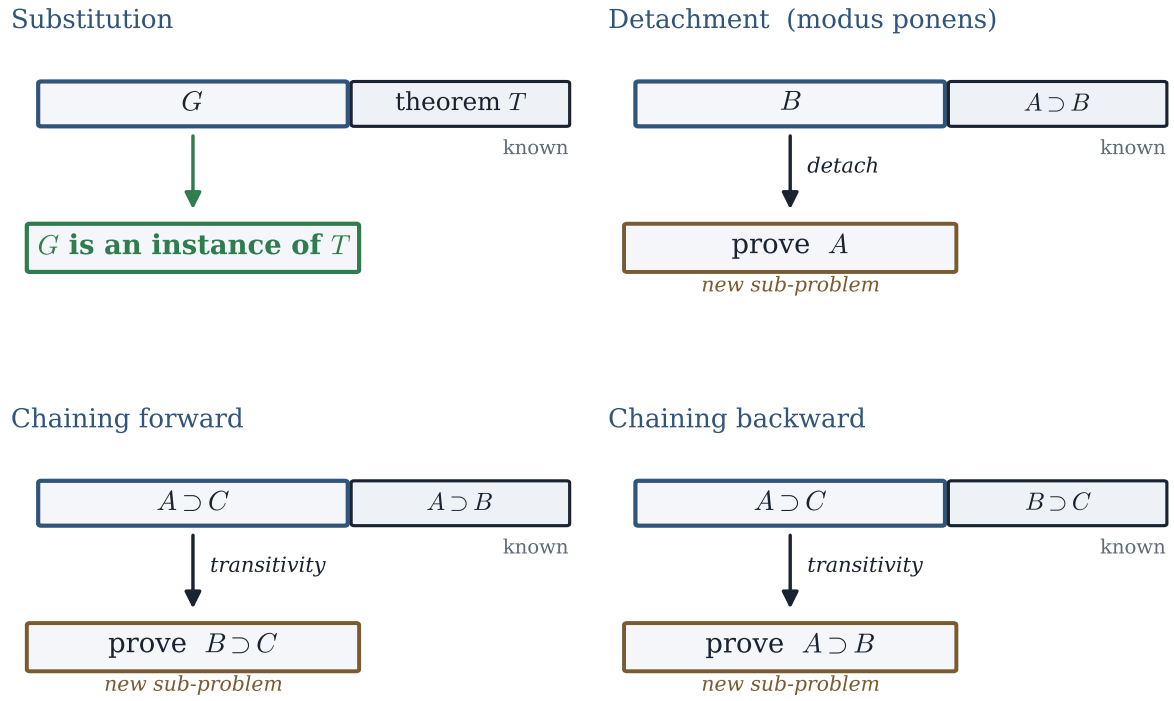



**Figure 1.2:** The four methods. SUBSTITUTION closes a goal directly; the other three reduce a goal $G$ to a single new subproblem, using a known axiom or theorem as the pivot.

**Substitution.** Search the known axioms and theorems for one that can be turned into the goal by substitution (and definitional replacement). If the goal is an *instance* of something already established, it is proved, with no subproblem generated. This is the only method that can terminate a line of proof.

**Detachment.** To prove $B$, look for a known theorem of the form $A \supset B$ (matching its consequent to $B$). If one is found, then by detachment it suffices to prove its antecedent $A$; so $A$ becomes a new subproblem.

**Chaining forward.** When the goal itself is an implication $A \supset C$, look for a known theorem $A \supset B$ sharing the antecedent. Because implication is transitive, it then suffices to prove $B \supset C$, which becomes the subproblem.

**Chaining backward.** Again for a goal $A \supset C$, look instead for a known $B \supset C$ sharing the *consequent*; it then suffices to prove $A \supset B$.

Two features of this repertoire are worth emphasizing. First, the methods are *goal-directed*: they reason from what is to be shown toward what is known, rather than blindly generating consequences of the axioms. Second, they are a deliberately *partial* selection of PM's derivational patterns—detachment and the transitivity of implication are ubiquitous in PM's own proofs. LT did not attempt every legal move; it attempted a few humanly natural ones. This is the sense in which its methods are heuristic rather than complete: there are valid theorems no combination of these four methods will reach, and LT indeed failed on 14 of the 52 theorems it attempted.

## 1.6 The task environment: the theorems of Chapter 2

LT was set to work on the first 52 theorems of Chapter 2 of *Principia Mathematica*, attempted in roughly PM's own numbering. Table 1.1 lists a representative sample. The sequence is pedagogically graded: the earliest theorems fall out of a single axiom by substitution or one detachment, while later ones require several reductions and the reuse of intermediate results. Because LT added every theorem it proved to its memory, the effective difficulty of a given theorem depended on how much of the chapter had already been established when it was attempted—so that, as in PM itself, the order of presentation is part of the mathematics. It is worth noting that even so elementary a proposition as the law of identity $p \supset p$ ($*2.08$) is not an axiom and must be derived, typically by chaining an instance of $*1.3$ (Add) with an instance of $*1.2$ (Taut) through the transitivity of implication.

**Table 1.1:** A representative sample of the Chapter-2 theorems LT attempted, in PM's numbering. LT proved them cumulatively, each becoming available for the proofs of those that follow.

| No. | Statement | Name / role |
|---|---|---|
| $*2.01$ | $(p \supset \neg p) \supset \neg p$ | *reductio ad absurdum* |
| $*2.02$ | $q \supset (p \supset q)$ | |
| $*2.03$ | $(p \supset \neg q) \supset (q \supset \neg p)$ | transposition |
| $*2.05$ | $(q \supset r) \supset \big((p \supset q) \supset (p \supset r)\big)$ | syllogism |
| $*2.06$ | $(p \supset q) \supset \big((q \supset r) \supset (p \supset r)\big)$ | syllogism |
| $*2.08$ | $p \supset p$ | identity |
| $*2.15$ | $(\neg p \supset q) \supset (\neg q \supset p)$ | transposition |
| $*2.17$ | $(\neg q \supset \neg p) \supset (p \supset q)$ | transposition |
| $*2.21$ | $\neg p \supset (p \supset q)$ | |
| $*2.85$ | $\big((p \vee q) \supset (p \vee r)\big) \supset \big(p \vee (q \supset r)\big)$ | improved by LT |

## 1.7 Executive control and heuristic search

The methods generate a *tree* of subproblems, and some discipline is needed to explore it without drowning in possibilities. This was the job of the *executive routine* (Figure 1.3). Presented with a goal, the executive first tried the substitution method against the store of known results; if that failed it applied detachment and the two chaining methods, each of which might spawn a subproblem, and it then recursed on those subproblems. Crucially, the store of "known results"

grew as work proceeded: LT attempted the theorems of Chapter 2 roughly in PM's own order, and each theorem it proved was added to memory and made available for the proofs of later, harder theorems—exactly the cumulative structure by which PM itself is built.

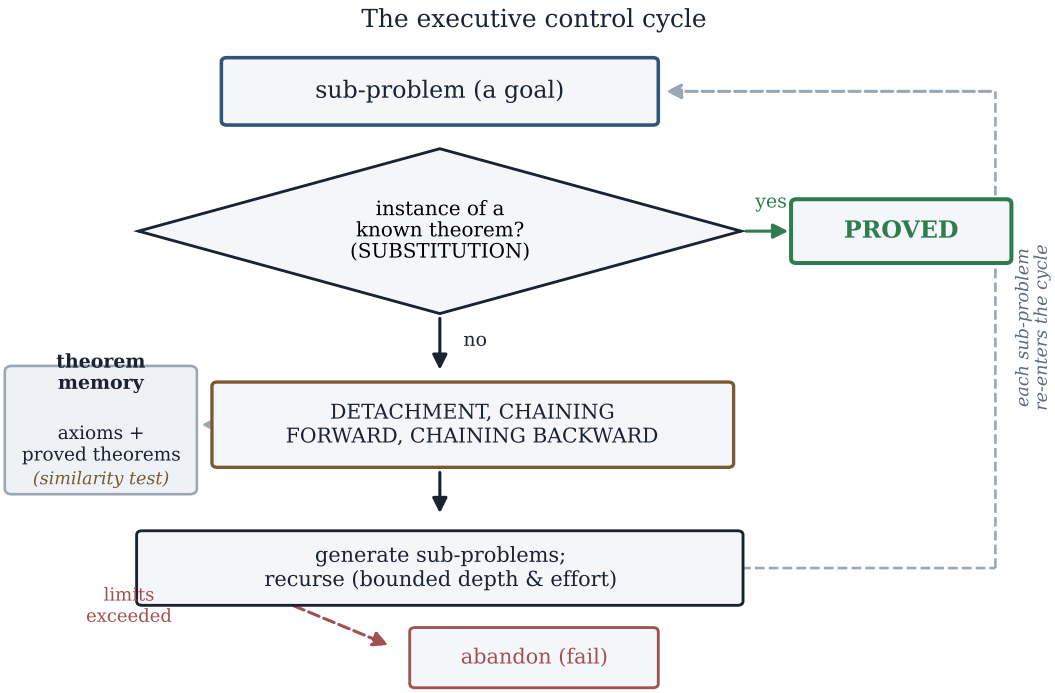



**Figure 1.3:** The executive control cycle. Each subproblem is first tested against memory by substitution; failing that, the reduction methods are applied under the similarity test, and their subproblems re-enter the cycle under bounds on depth and effort.

The central difficulty is combinatorial. Even in so small a language, the number of substitutions one might make, and of known theorems one might try to detach or chain, explodes rapidly with depth. Newell, Shaw, and Simon contrasted their approach with what they called the "British Museum algorithm"—the exhaustive generation of all provable formulas in order of length—which, though it would eventually produce any given theorem, is hopeless in practice. LT's answer was heuristic pruning, and its principal instrument was the *similarity test*. Before investing effort in matching a candidate theorem against a goal, LT compared their cheap descriptions—counts of variables, of levels of nesting, and the like—and abandoned the attempt at once if the descriptions were too dissimilar for a match to be possible. A great many futile lines were never entered because a constant-time comparison ruled them out. The term *heuristic*, in the technical sense it now carries in artificial intelligence, enters the field here; Newell had absorbed the notion from Polya's writing on mathematical discovery, and LT gave it a mechanical realization.

The necessity of such pruning can be felt quantitatively. The rule of substitution allows *any* formula to be put for a variable, so the set of instances of even a single axiom is infinite, and the number of ways to combine a growing store of theorems by detachment and chaining rises steeply with the depth of search. The "British Museum" strategy of generating all provable formulas in order of length would, long before reaching a target of quite modest size, produce an astronomical quantity of irrelevant well-formed formulas; it is correct in the limit and useless in practice. Heuristic selection is therefore not an optimization bolted onto a workable method but the thing that makes the method workable at all.

The similarity test exploits a simple structural fact. Substitution can only make an expression *larger*: it can add variables and deepen nesting, but it can never merge two distinct variables into one or flatten structure away. A known theorem $T$ can therefore be specialized to a goal $G$ only if $G$ is at least as rich as $T$ on the features recorded in their descriptions—at least as many distinct variables, at least as deep. Comparing the two descriptions is a constant-time operation, and whenever it fails LT can discard the pairing of $T$ with $G$ without attempting the far more expensive structural match.

Two further disciplines bounded the search. LT worked under explicit limits on effort and on memory—the JOHNNIAC's core was small—abandoning subproblems that grew too large or lines that ran too long. And because the reduction methods each produce a *simpler* subproblem (a shorter implication, or an antecedent), the subproblem tree tends toward the axioms rather than away from them. The result is a search that is neither exhaustive nor guaranteed, but that, guided by structural resemblance to known results, finds proofs of the kind a person might find, and sometimes finds them quickly.

## 1.8 A worked example: theorem ∗2.85

The theorem on which LT is most famous for outdoing its source is ∗2.85,

$$\big((p \vee q) \supset (p \vee r)\big) \;\supset\; \big(p \vee (q \supset r)\big), \tag{1.1}$$

a form of the distribution of disjunction over implication. Figure 1.4 shows a proof of it found by the reconstruction, and it illustrates the machinery of the preceding sections in miniature.

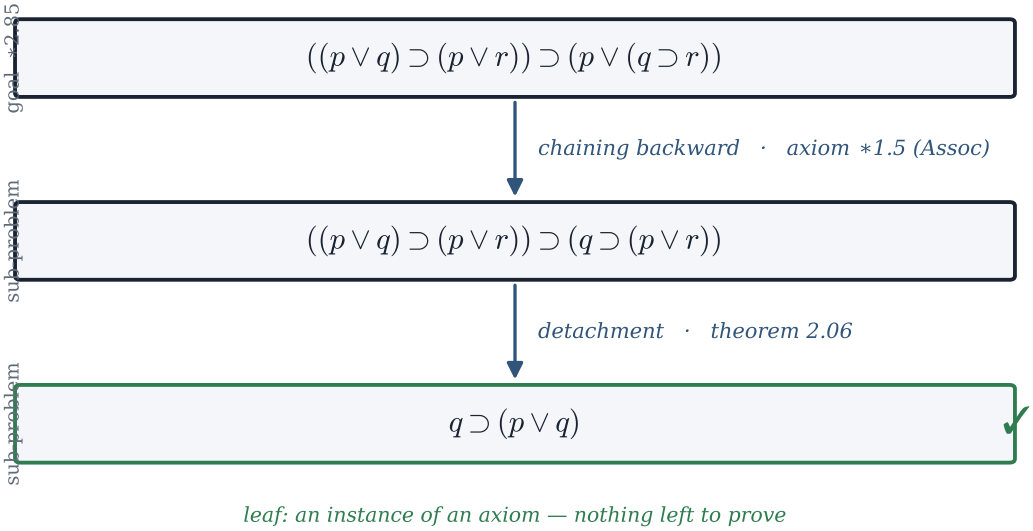



**Figure 1.4:** A proof of ∗2.85 found by the reconstruction. Reading downward, the goal is reduced by chaining backward, then by detachment against a theorem proved earlier in the run, to a subproblem that is a direct instance of an axiom.

The executive first treats the goal, an implication, by *chaining backward* through axiom ∗1.5 (Assoc), reducing it to the subproblem $\big((p \vee q) \supset (p \vee r)\big) \supset \big(q \supset (p \vee r)\big)$. That subproblem is then handled by *detachment* against theorem 2.06—itself proved earlier in the same run and remembered—which leaves the antecedent $q \supset (p \vee q)$ to be shown. And that is nothing other than axiom ∗1.3 (Add) with its variables suitably filled in: the *substitution* method closes it, and the proof is complete. The derivation is short because the search is free to assemble

whatever combination of remembered results reaches the goal; it is not constrained to retrace the particular route PM's authors took.

Historically, LT's proof of ∗2.85 was shorter than the proof printed in *Principia Mathematica*, a fact its authors noted with some pride. Simon showed the machine's proof to Bertrand Russell, who by one account responded with delight. When Newell and Simon submitted a note on the result to the *Journal of Symbolic Logic*, it was declined as a fresh proof of an already-known elementary theorem—the referees, the story goes, not remarking that one of its authors was a program.

## 1.9 Relationship to *Principia Mathematica*

LT's relationship to PM is a study in inheritance and inversion. What it *inherited* was nearly everything concrete: the object language of negation and disjunction with defined implication; the five primitive propositions verbatim; the rules of substitution and detachment; and, as its actual problem set, the numbered theorems of Chapter 2 in something close to their original order, proved cumulatively so that earlier results support later ones. In a quite literal sense LT's task was to reconstruct a stretch of PM from the inside.

What it *inverted* was the point of the exercise. *Principia Mathematica* is a work of foundations: its proofs are justifications, links in a chain meant to secure arithmetic upon logic, and their virtue is rigor and completeness, not the manner of their discovery. The formal system is a static object, and a theorem's proof is a timeless certificate that it belongs. LT, by contrast, was never interested in logic for its own sake—by the mid-1950s the ramified apparatus of PM was, as Simon observed, outmoded for foundational purposes—but in *how proofs are found*. Its authors were building a theory of human problem solving and needed a well-defined arena in which to test it; PM's propositional calculus supplied one. The proofs LT produced were therefore not primarily justificatory but *investigative*: traces of a search, artifacts of a process that could equally have failed, and indeed did fail on many theorems.

This inversion has an epistemic edge. Post's theorem guarantees that PM's propositional calculus is complete—every tautology has a proof—so a mechanical procedure could in principle certify any propositional theorem whatever. LT pointedly declined that route. It accepted incompleteness in its method in exchange for search that was tractable and, its authors argued, psychologically faithful. The contrast is thus not between two proof systems but between two attitudes toward proof: PM asks whether a theorem *is* derivable and, being complete, answers yes for all tautologies; LT asks how a reasoner might *come to derive* one, and answers only sometimes, in the way that people do. Even the redundancy Bernays identified in PM's axioms—the superfluity of ∗1.5—is invisible from LT's standpoint, which cares not about the economy of the axiom set but about the findability of proofs. The ∗2.85 episode is the emblem of the whole relationship: the pupil program, working within the master's system, found a route its authors had missed, precisely because it was searching rather than transcribing.

## 1.10 Significance, limitations, and legacy

Considered as an artifact, LT is narrow: it addressed only the propositional fragment of PM, proved fewer than three-quarters of the theorems it attempted, and was bounded at every turn by the tiny memory of its machine. Its incompleteness was not incidental but constitutive—the price of its heuristic character. It is easy, in retrospect, to see all the things it could not do.

Considered as a source of ideas, however, its influence is difficult to overstate. Several strands begin here. The first is *heuristic search*: the treatment of problem solving as the exploration of a

space of states or subproblems, guided by inexpensive rules of thumb that prune away the futile majority of options. This conception, generalized from LT's methods and similarity test, became a organizing principle of artificial intelligence. The second is *list processing*: IPL, devised to give LT its flexible symbolic memory, established the linked list as the natural medium of symbolic computation and influenced LISP, in which the field's symbolic tradition would be carried on. The third is *cognitive simulation*. LT was intended from the outset as a theory of human thinking rendered precise enough to run, and it became a template for the information-processing psychology that Newell and Simon went on to build—most immediately in the General Problem Solver (1957), which generalized LT's goal-directed reductions into the technique of means–ends analysis, and eventually in a research program that helped launch the cognitive revolution. That LT was meant as psychology and not merely as engineering is not a retrospective gloss: Newell and Simon compared the machine's behavior with the think-aloud protocols of human subjects working the same problems, treating agreement and disagreement between program and person as evidence about a theory of mind. This move—an executable program offered as a precise, testable psychological theory—was as consequential for cognitive science as the search techniques were for artificial intelligence.

There is, finally, a reason LT still repays study beyond its historical priority. It states, in unusually clean form, a distinction that remains central: between verifying that a conclusion follows and *discovering* that it does. *Principia Mathematica* perfected the first; the Logic Theorist inaugurated the machine pursuit of the second. The two live together in every automated reasoner since.

## 1.11 Conclusion

The Logic Theorist took the propositional calculus of *Principia Mathematica*—its language, its five axioms, its rules, and its very theorems—and used it as the arena for a new kind of question. Representing formulas as list structures, applying substitution, replacement, and detachment through four goal-directed methods, and holding the combinatorial explosion at bay with a similarity heuristic under an executive that learned as it went, it found proofs in the manner of a searching mathematician rather than a grinding decision procedure. In doing so it inherited PM's system whole while reversing its purpose, trading the completeness of foundational logic for the tractability and human resemblance of heuristic discovery—and, in the proof of $*2.85$, quietly improving on the monument it had set out merely to imitate.

# Chapter 2

# The Theory of Deduction (∗1–∗5)

*This is the first installment of a commentary on the abridged* Principia Mathematica to ∗56*—the logical core of Whitehead and Russell's system. It treats the propositional calculus, numbers* ∗1 *through* ∗5*: the primitive ideas and axioms of the theory of deduction, the first cascade of derived theorems, and the development of conjunction, equivalence, and the propositional "algebra." The commentary is deliberately hybrid. Each number opens with motivation—what the authors are trying to secure and why it comes where it does—followed by proposition-by-proposition remarks that restate the content in modern notation, explain the proof idea, and connect it to later logic. Because this stretch of* Principia Mathematica *is purely truth-functional, every theorem here can be, and has been, verified mechanically; where that illuminates the text we say so.*

## 2.1 General introduction

### 2.1.1 The aim, in one paragraph

*Principia Mathematica* (1910–13) is the great monument of *logicism*, the thesis that mathematics is a part of logic—that its concepts are definable from logical concepts and its theorems provable from logical axioms. The three volumes attempt to make good on that thesis by actually carrying out the reduction, building arithmetic, the theory of relations, and much of analysis on a base of pure logic, in an explicit formal notation, with proofs written out in full. Whether logicism *succeeds* is a question the book cannot settle in its own terms—the axioms of infinity and reducibility, in particular, look less like logic the harder one stares at them—but the sheer fact of the construction changed what a foundation for mathematics could look like.

### 2.1.2 Why the propositional calculus, and why first

Everything in *Principia Mathematica* rests on the theory of deduction: the logic of *propositions* joined by "not," "or," "if–then," "and," and "if and only if." Before one can reason about classes, relations, or numbers, one needs the inferential glue that holds any proof together, and that glue is the propositional calculus of ∗1–∗5. These numbers are also, by a wide margin, the most accessible part of the book: they involve no quantifiers, no types, no paradoxes—just five axioms and a rule of inference, from which a long list of familiar logical laws is extracted. This is the part later reconstructed by hand and by machine, from Newell, Shaw, and Simon's Logic Theorist in 1956 to the proof-assistant formalizations of today, and it is the natural place for a commentary to begin.

### 2.1.3 What "to $*56$" contains

The abridged volume runs through the logical heart of Volume I. In outline: the theory of deduction ($*1$–$*5$, propositional logic); the theory of apparent variables ($*9$–$*14$: quantification, the ramified theory of types and the axiom of reducibility, identity, and the celebrated theory of definite descriptions); classes and relations ($*20$–$*43$); and the first steps toward cardinal arithmetic, on unit classes and couples ($*50$–$*56$). The present installment covers only the first block. Later installments will take up quantification and descriptions ($*9$–$*14$), then classes and relations.

### 2.1.4 How to read this commentary

The register is a hybrid of reader's guide and scholarly commentary. A modern reader who knows elementary logic should be able to use it to get *through Principia Mathematica*; a more specialized reader should find enough about proof strategy, historical context, and the connections to later logic to make it worth the shelf. Two standing conventions. First, we give each proposition in modern notation, reserving *Principia Mathematica*'s own dot notation for the primer below and occasional illustration. Second, because $*1$–$*5$ are truth-functional, we treat semantic validity (being a tautology) as a check on our reading: every formula we attribute to *Principia Mathematica* in this installment has been verified by truth table, and we occasionally note when a theorem's mechanical proof is instructive.

## 2.2 A note on notation and framework

**Primitive ideas.** *Principia Mathematica*'s propositional calculus takes as primitive the notion of an *elementary proposition* and two connectives: negation $\neg p$ ("not $p$") and disjunction $p \vee q$ ("$p$ or $q$," inclusive). Everything else in $*1$–$*5$ is *defined*. Implication is the first and most important definition,

$$p \supset q \quad \stackrel{\text{def}}{=} \quad \neg p \vee q, \tag{$*1.01$}$$

so *Principia Mathematica*'s "if–then" is material implication from the outset. Conjunction and equivalence come later, in $*3$ and $*4$, by further definitions.

**The assertion sign.** A line of *Principia Mathematica* asserting a theorem begins with Frege's turnstile $\vdash$, read "it is true that." The sign is not part of the object language; it marks the transition from displaying a proposition to *asserting* it. We generally omit it, since our modern formulas are understood as assertions.

**The dot notation.** *Principia Mathematica* has no parentheses. Grouping is shown by *dots*, whose number indicates scope: a connective flanked by more dots binds more loosely than one flanked by fewer. Thus what we would write $(p \vee q) \supset (q \vee p)$ appears as $p \vee q \,.\supset.\, q \vee p$, the double dot around $\supset$ making it the main connective. The system is elegant and maddening in equal measure, and its opacity is one reason *Principia Mathematica* is more often cited than read; we translate it away.

**Definitions and primitive propositions.** A line tagged *Df* is a definition, introducing shorthand; a line tagged Pp ("primitive proposition") is an axiom or a primitive rule, asserted without proof. The theory of deduction has exactly one primitive rule of *inference* and a short list of primitive *propositions*.

**The rules.** Two rules move from asserted lines to new ones. The first is *modus ponens*, or detachment ($*1.1$): from $p$ and $p \supset q$, assert $q$. The second, used constantly but stated only informally, is *substitution*: in an asserted proposition one may replace a propositional variable, throughout, by any proposition. A third, bookkeeping operation is *replacement by definition*: anything defined may be swapped for its definiens, so that $p \supset q$ and $\neg p \vee q$ are interchangeable wherever they occur. These three—detachment, substitution, and definitional replacement—are the whole engine of $*1$–$*5$.

**Modern notation used here.** $\neg$ negation, $\vee$ disjunction, $\wedge$ conjunction (*Principia Mathematica*'s dot), $\supset$ material implication, $\equiv$ material equivalence (*Principia Mathematica*'s $\equiv$). Variables $p, q, r, s$ range over propositions.

## 2.3 $*1$ Primitive ideas and primitive propositions

**Motivation.** Number $*1$ lays the foundation: it fixes the primitive vocabulary, defines implication, states the one rule of inference, and posits the axioms from which the entire propositional calculus—and, mediately, the whole book—will flow. Everything is chosen for deductive power, not economy or elegance; the authors want a base ample enough to make the derivations of $*2$–$*5$ go through smoothly.

After the definition $*1.01$ and the rule of detachment $*1.1$ (with its companion $*1.11$ governing propositional functions), *Principia Mathematica* asserts five primitive propositions. In modern notation:

| | | |
|---|---|---|
| $*1.2$ | (Taut) | $(p \vee p) \supset p$ |
| $*1.3$ | (Add) | $q \supset (p \vee q)$ |
| $*1.4$ | (Perm) | $(p \vee q) \supset (q \vee p)$ |
| $*1.5$ | (Assoc) | $\big(p \vee (q \vee r)\big) \supset \big(q \vee (p \vee r)\big)$ |
| $*1.6$ | (Sum) | $(q \supset r) \supset \big((p \vee q) \supset (p \vee r)\big)$ |

Three further primitive propositions, $*1.7$, $*1.71$, $*1.72$, are closure conditions: the negation of an elementary proposition is elementary, the disjunction of two elementary propositions is elementary, and likewise for elementary functions. They ensure the primitive operations keep us inside the intended range and play no role in ordinary derivations.

**Remarks.** Three points repay attention. First, *the axioms are stated with a defined sign*: $\supset$ appears in $*1.2$–$*1.6$ although it is not primitive, a harmless but revealing inheritance from Russell's earlier practice—the disjunctive primitives are doing the real work underneath. Second, *the set is not independent.* Bernays showed in 1926 that $*1.5$ (Assoc) is derivable from the other four; *Principia Mathematica*'s basis is redundant, a fact its authors did not know and would not have minded, since findability of proofs, not axiomatic economy, was their concern. Third, *the system is complete.* Post proved in 1921 that the theorems of this calculus are exactly the tautologies: every truth-functional validity is derivable. That theorem is the backdrop against which everything in $*2$–$*5$ is a foregone conclusion *semantically*, and yet a real achievement *syntactically*—the interest lies in the derivations, not the truths.

## 2.4 $*2$ Immediate consequences of the primitive propositions

**Motivation.** With the axioms in place, $*2$ begins to *deduce.* Its business is to extract from five terse axioms the stock of everyday logical laws—identity, the syllogism, excluded middle, double negation, transposition, reductio—that any working logic must have. Read as a whole,

$*2$ is a demonstration that the meagre basis of $*1$ is in fact sufficient: the familiar principles are not further assumptions but consequences. It is also the number on which every later proof leans most heavily, and the one the Logic Theorist took as its proving ground.

**Identity and the syllogism.** The first landmark is the law of identity,

$*\mathbf{2.08}$ $p \supset p$,

which, though utterly obvious, is *not* an axiom and must be earned—typically by chaining an instance of Add ($*1.3$) with Taut ($*1.2$) through the transitivity of implication. Transitivity itself appears as the two forms of the hypothetical syllogism,

$*\mathbf{2.05}$ $(q \supset r) \supset \big((p \supset q) \supset (p \supset r)\big)$,

$*\mathbf{2.06}$ $(p \supset q) \supset \big((q \supset r) \supset (p \supset r)\big)$,

which differ only in the order of the premises and each say that implication chains. These are the workhorses of the entire book: almost every later proof is, at bottom, a syllogistic chain licensed by $*2.05$ or $*2.06$.

**The "paradoxes" of material implication.** Because $\supset$ is material, $*2$ proves several principles that are valid but jar the ear:

$*\mathbf{2.02}$ $q \supset (p \supset q)$,

$*\mathbf{2.21}$ $\neg p \supset (p \supset q)$,

$*\mathbf{2.24}$ $p \supset (\neg p \supset q)$.

A true proposition is implied by anything ($*2.02$); a false one implies anything ($*2.21$, *ex falso quodlibet*). These are not defects in the proofs but consequences of the decision at $*1.01$ to read "if" truth-functionally, and they are worth flagging to a new reader precisely because they are where the material conditional first bites.

**Negation: excluded middle and double negation.** Number $*2$ secures the classical laws of negation,

$*\mathbf{2.11}$ $p \vee \neg p$,

$*\mathbf{2.12}$ $p \supset \neg\neg p$,

$*\mathbf{2.14}$ $\neg\neg p \supset p$,

the last two together giving the equivalence of $p$ with its double negation—the step that makes the logic classical rather than intuitionistic. (*Principia Mathematica* has no qualms here: double negation is proved in both directions and used freely.)

**Transposition.** The contraposition laws arrive as a family,

$*\mathbf{2.16}$ $(p \supset q) \supset (\neg q \supset \neg p)$,

$*\mathbf{2.17}$ $(\neg q \supset \neg p) \supset (p \supset q)$,

with $*2.15$ and $*2.03$ giving the mixed forms. Together they let one flip an implication and negate, a move used constantly in later reductio arguments.

**Reductio and *consequentia mirabilis*.** Two striking principles close the important part of $*2$:

$*\mathbf{2.01}$ $(p \supset \neg p) \supset \neg p$,

$*\mathbf{2.18}$ $(\neg p \supset p) \supset p$.

The first is the principle of *reductio ad absurdum* in its purest form—if $p$ implies its own negation, then $p$ is false. The second, sometimes called the *consequentia mirabilis* or Clavius' law, is its dual and is genuinely surprising: if the falsehood of $p$ would imply its truth, then $p$ is true. It is the propositional seed of proof by contradiction.

**The famous $*2.85$.** The number ends with a cluster of "sum" and "product" laws for disjunction and implication, of which the most celebrated is

$*\mathbf{2.85}$ $\big((p \vee q) \supset (p \vee r)\big) \supset \big(p \vee (q \supset r)\big)$.

This is the theorem for which the Logic Theorist found a proof shorter than *Principia Mathematica*'s own—Simon showed it to Russell, who was delighted—and it is a good emblem of the whole number: an unobvious validity reached by combining, through detachment and chaining, laws already proved. A reader meeting $*2$ for the first time can safely treat its seventy-odd propositions as a toolbox and learn the tools above; the rest are variants and combinations.

## 2.5 $*3$ The logical product (conjunction)

**Motivation.** Number $*3$ introduces conjunction and develops its laws. The point is not novelty—conjunction is definable and adds no expressive power—but *convenience*: proofs about "$p$ and $q$" are ubiquitous, and *Principia Mathematica* needs the rules that let premises be combined, split, and shuffled. The number is best read as building the algebra of "and" and its interaction with implication.

*Principia Mathematica* defines the product from the primitives by De Morgan's recipe,

$$p \wedge q \quad \stackrel{\text{def}}{=} \quad \neg(\neg p \vee \neg q), \tag{$*3.01$}$$

so that "$p$ and $q$" just abbreviates "not (not-$p$ or not-$q$)." The essential laws follow.

**Composition and simplification.** One builds a conjunction and takes it apart:

$*\mathbf{2.3\ /\ 3.2}$ $p \supset \big(q \supset (p \wedge q)\big)$,

$*\mathbf{3.26}$ $(p \wedge q) \supset p$,

$*\mathbf{3.27}$ $(p \wedge q) \supset q$.

$*3.2$ says that from $p$ and (separately) $q$ one may assert their conjunction; $*3.26$ and $*3.27$ ("simplification") say a conjunction yields each conjunct. These three are used on nearly every subsequent page.

**Importation and exportation.** The hinge between conjunction and iterated implication:

$*\mathbf{3.3}$ $\big((p \wedge q) \supset r\big) \supset \big(p \supset (q \supset r)\big)$,

$*\mathbf{3.31}$ $\big(p \supset (q \supset r)\big) \supset \big((p \wedge q) \supset r\big)$.

Exportation ($*3.3$) and importation ($*3.31$) together say that "$p$ and $q$ together imply $r$" is interchangeable with "$p$ implies that $q$ implies $r$"—the formal counterpart of discharging premises one at a time. This equivalence is what makes multi-premise reasoning tractable in a system whose only rule is detachment.

**Assertion, and composition of implications.** A theorem-form of modus ponens appears as

$*\mathbf{3.35}$ $\big(p \wedge (p \supset q)\big) \supset q$,

and the "composition" laws let parallel implications be combined,

$*\mathbf{3.43}$ $\big((p \supset q) \wedge (p \supset r)\big) \supset \big(p \supset (q \wedge r)\big)$,

with $*3.45$ ("factor") and $*3.47$ extending the pattern to conjunctions of antecedents and consequents. The flavour of $*3$ is exactly this: turning the handling of "and" into reliable, citable machinery.

## 2.6 $*4$ Equivalence and the algebra of propositions

**Motivation.** Number $*4$ introduces material equivalence and, with it, turns the propositional calculus into a genuine *algebra*. Once "$p \equiv q$" is available and shown to behave like an equality—reflexive, symmetric, transitive—one can *rewrite* propositions by substituting equivalents, and the classical identities (double negation, De Morgan, distribution, commutation, association) become the transformation rules of that algebra. This is the number that makes *Principia Mathematica*'s propositional logic feel like calculation.

Equivalence is defined as mutual implication,

$$p \equiv q \quad \stackrel{\text{def}}{=} \quad (p \supset q) \wedge (q \supset p). \tag{$*4.01$}$$

**Equivalence is an equivalence relation.** *Principia Mathematica* proves

$*\mathbf{4.2}$ $p \equiv p$,

$*\mathbf{4.21}$ $(p \equiv q) \equiv (q \equiv p)$,

$*\mathbf{4.22}$ $\big((p \equiv q) \wedge (q \equiv r)\big) \supset (p \equiv r)$,

i.e. reflexivity, symmetry, and transitivity. These three are quietly foundational: they license the substitution of equivalents that the rest of the number, and much of the book, relies on.

**Double negation and transposition.** $*\mathbf{4.13}$ $p \equiv \neg\neg p$,

$*\mathbf{4.1}$ $(p \supset q) \equiv (\neg q \supset \neg p)$.

Double negation now appears as a full equivalence (sharpening $*2.12/*2.14$), and transposition is likewise upgraded to an equivalence.

**The "laws of tautology," commutation, association.** The idempotence laws, which Russell calls the principles of tautology,

$*\mathbf{4.24}$ $p \equiv (p \wedge p)$,

$*\mathbf{4.25}$ $p \equiv (p \vee p)$,

say that repetition is idle; commutativity ($*4.3$: $p \wedge q \equiv q \wedge p$; $*4.31$: $p \vee q \equiv q \vee p$) and the associative laws that follow complete the familiar structure.

**De Morgan and distribution.** The high points of the number are the duality and distributive laws,

$*\mathbf{4.51}\quad \neg(p \wedge q) \equiv (\neg p \vee \neg q),$

$*\mathbf{4.4}\quad p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r),$

$*\mathbf{4.41}\quad p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r),$

together with the companion De Morgan law $*4.5$ and the relations

$*\mathbf{4.6}\quad (p \supset q) \equiv (\neg p \vee q)$

tying implication back to disjunction. With these, any propositional formula can be pushed toward a normal form—the practical pay-off of the whole number.

## 2.7 $*5$ Miscellaneous propositions

**Motivation.** Number $*5$ is, by its own title, a miscellany: it gathers further propositional laws—mostly about the interaction of product, sum, and equivalence—that will be wanted later but did not fit the tidy development of $*3$–$*4$. It is the shortest and least systematic of the five, and a reader may treat it as a reference appendix to the propositional calculus rather than a narrative. A few of its results are worth singling out.

$*\mathbf{5.1}\quad (p \wedge q) \supset (p \equiv q).$

If two propositions are both true, they are (materially) equivalent—a small law with outsized use in later equivalence-chasing.

$*\mathbf{5.32}\quad \big(p \supset (q \equiv r)\big) \equiv \big((p \wedge q) \equiv (p \wedge r)\big).$

A conditional equivalence can be traded for an equivalence of conjunctions; this is the form in which "$\equiv$" is most often manipulated under hypotheses.

$*\mathbf{5.6}\quad \big((p \wedge \neg q) \supset r\big) \equiv \big(p \supset (q \vee r)\big).$

A case-splitting law relating a conjunctive antecedent to a disjunctive consequent, useful whenever a proof proceeds by "either $q$, or else ...".

Two further curiosities close the number's logical interest. *Principia Mathematica* proves

$*\mathbf{5.13}\quad (p \supset q) \vee (q \supset p),$

that of any two propositions one implies the other—valid classically, and a small shock to intuition—and the near-relatives $*5.11$ and $*5.12$. With $*5$ the propositional calculus is complete as a working tool; from $*6$ onward *Principia Mathematica* turns to quantification.

## 2.8 Editorial notes and what comes next

The propositional numbers are the firmest ground in *Principia Mathematica*, and the part a commentary can most fully underwrite: because $*1$–$*5$ are truth-functional, every assertion in them is mechanically checkable, and the Appendix records that check for each proposition cited above. The difficulty rises sharply at $*9$. There the quantifiers enter, and with them the ramified theory of types, the axiom of reducibility, and the two rival treatments of quantification ($*8$ versus $*9$) that mark the difference between the first and second editions—material where proof strategy and philosophical commitment become entangled and where a commentary must lean on the primary text and the secondary literature rather than on truth tables. Installment II will take up $*9$–$*14$: the theory of apparent variables, identity, and the theory of definite descriptions, including $*14$'s famous analysis of "the."

## 2.9 Machine verification

Every propositional formula attributed to *Principia Mathematica* in this installment was translated into the primitive vocabulary ($\neg$, $\vee$, with $\supset$, $\wedge$, $\equiv$ expanded by their definitions) and evaluated on all truth-value assignments. Each of the following came out a tautology, as a theorem of the classical propositional calculus must:

> $*1.2$, $*1.3$, $*1.4$, $*1.5$, $*1.6$; $*2.01$, $*2.02$, $*2.03$, $*2.04$, $*2.05$, $*2.06$, $*2.08$, $*2.11$, $*2.12$, $*2.13$, $*2.14$, $*2.15$, $*2.16$, $*2.17$, $*2.18$, $*2.21$, $*2.24$, $*2.25$, $*2.27$, $*2.85$; $*3.2$, $*3.26$, $*3.27$, $*3.3$, $*3.31$, $*3.35$, $*3.43$, $*3.45$, $*3.47$; $*4.1$, $*4.11$, $*4.13$, $*4.2$, $*4.21$, $*4.22$, $*4.24$, $*4.25$, $*4.3$, $*4.31$, $*4.4$, $*4.41$, $*4.5$, $*4.51$, $*4.6$, $*4.7$; $*5.1$, $*5.11$, $*5.12$, $*5.13$, $*5.32$, $*5.6$.

The definitions $*1.01$ ($\supset$), $*3.01$ ($\wedge$), and $*4.01$ ($\equiv$) are not theorems but stipulations, and are used to carry out the translation. The five primitive propositions $*1.2$–$*1.6$ are of course also tautologies, as the completeness of the calculus (Post, 1921) requires.

# Chapter 3

# Quantification, Types, Identity, and Descriptions (∗9–∗14)

*This second installment covers the numbers in which* Principia Mathematica *passes from the logic of propositions to the logic of* propositional functions*: quantification over one and several variables (∗9–∗11), the theory of types and the axiom of reducibility (∗12), the definition of identity (∗13), and the theory of definite descriptions (∗14). This is where* Principia Mathematica *becomes philosophically deep and technically contentious—where the quantifiers, the ramified type hierarchy, and the notorious axiom of reducibility enter, and where Russell's celebrated analysis of "the" is carried out. As before, each number opens with motivation and continues with commentary on its pivotal propositions in modern notation, now with more attention to interpretation, since here proof strategy and philosophical commitment are entangled.*

## 3.1 Introduction: from propositions to propositional functions

Installment I concerned whole propositions combined by connectives. The numbers now before us make the decisive move to *propositional functions*—expressions like $\phi x$ ("$x$ is $\phi$") containing a variable—and to the *quantifiers* that bind them: $(x).\,\phi x$ ("all $x$ are $\phi$") and $(\exists x).\,\phi x$ ("some $x$ is $\phi$"). With quantification the system acquires the expressive power of modern predicate logic, and ∗10–∗11 are, in effect, *Principia Mathematica*'s predicate calculus.

But quantification over functions immediately raises the problem that had nearly wrecked Russell's project a decade earlier—the paradoxes. If a function may take itself as argument, one can form "the function true of just those functions not true of themselves" and reproduce the contradiction. *Principia Mathematica*'s response is the *ramified theory of types* (∗12), which stratifies functions by both type and order and forbids the offending self-application. The cure, however, is almost too strong: ramification blocks not only the paradoxes but also ordinary mathematics, which is full of harmless "impredicative" definitions (the least upper bound of a bounded set is defined by quantifying over the very set of upper bounds it belongs to). To restore mathematics without abandoning the hierarchy, *Principia Mathematica* adds the *axiom of reducibility*—the single most criticized assumption in the book. Identity (∗13) is then defined in terms of predicative functions, and descriptions (∗14) receive Russell's famous contextual analysis.

**An edition warning.** A reader of the abridged *to* ∗56 meets two treatments of quantification. The first-edition development stands in the main text at ∗9; the second edition added, in Appendix A, an alternative ∗8 that reworks the theory of deduction for propositions with

apparent variables (in the spirit of the stroke-based, Wittgenstein-influenced second-edition Introduction). Both are printed in the abridged volume. We follow the main-text development (∗9, ∗10, ∗11) and remark on the ∗8 alternative at the close.

## 3.2 Notation and framework: what is new

**Propositional functions.** $\phi x, \psi x, \ldots$ are expressions containing a free variable that become propositions when the variable is fixed. *Principia Mathematica* is careful, and the modern reader must be too, to distinguish a function $\phi\hat{x}$ from its value $\phi x$ at an argument.

**Real vs. apparent variables.** A *real* (free) variable is one left open; an *apparent* (bound) variable is one governed by a quantifier. A standing convention of *Principia Mathematica*, easily missed, is that *asserting* a formula with a free variable amounts to asserting its universal closure: $\vdash \phi x$ has the force of $(x).\, \phi x$. Much of ∗9–∗10 is the careful management of this convention.

**Quantifier notation.** *Principia Mathematica* writes the universal quantifier $(x).\, \phi x$ and takes it (in ∗9) as primitive; the existential quantifier is defined,

$$(\exists x).\, \phi x \quad \stackrel{\text{def}}{=} \quad \neg\, (x).\, \neg \phi x. \tag{∗10.01}$$

A subscripted connective, as in $\phi x \equiv_x \psi x$, abbreviates a quantified conditional or biconditional: $(x).\, (\phi x \equiv \psi x)$.

**Predicative functions and the shriek.** To state the type restrictions *Principia Mathematica* marks *predicative* functions—those, roughly, of the lowest order compatible with their arguments, involving no apparent function-variables—with an exclamation point: $\phi!x$, read "$\phi$-shriek $x$." The distinction between $\phi x$ (any function) and $\phi!x$ (a predicative one) is exactly what the axiom of reducibility will bridge.

**The type hierarchy, in brief.** Individuals form the bottom type; functions of individuals the next; functions of those the next; and so on, with the rule that a function's argument must be of lower type. *Ramification* adds a second index, *order*, tracking the quantifiers a function's definition contains, so that a function may not be defined by quantifying over a totality to which it would itself belong (the vicious-circle principle). We keep the exposition informal; the working propositions of ∗9–∗11 can be read with types in the background.

**Descriptions.** The inverted iota $\imath x$ begins a definite description: $(\imath x)(\phi x)$ is "the $x$ such that $\phi x$," or "the $\phi$." The sign E! expresses that such a description is satisfied by exactly one thing. Both are explained in ∗14.

## 3.3 ∗9 The theory of apparent variables

**Motivation.** Number ∗9 does the foundational, unglamorous work of *extending the theory of deduction to quantified propositions.* The propositional calculus of ∗1–∗5 knows nothing of quantifiers; ∗9 must show how $\neg$ and $\vee$ behave when their arguments contain apparent variables, so that the whole apparatus of detachment and the primitive propositions carries over. It is the bridge from the propositional to the functional calculus.

The core is a set of definitions pushing negation and disjunction through quantifiers. Negation converts the quantifiers into one another,

$*\mathbf{9.01}\quad \neg\,(x).\,\phi x \stackrel{\text{def}}{=} (\exists x).\,\neg\phi x,$

$*\mathbf{9.02}\quad \neg\,(\exists x).\,\phi x \stackrel{\text{def}}{=} (x).\,\neg\phi x,$

and disjunction with a quantifier-free proposition is defined to distribute inward (*Principia Mathematica*'s $*9.03$–$*9.06$), e.g. $(x).\phi x\,.\vee\,.p$ is defined as $(x).\,(\phi x \vee p)$. On this basis $*9$ posits the primitive inferential facts about quantifiers—that an instance follows from a universal, and that an instance yields an existential,

$*\mathbf{9.1}\quad \phi x \supset (\exists z).\,\phi z,$

together with the rules needed to extend $*1$'s detachment to the new forms. The achievement of $*9$ is entirely structural: after it, one may reason with quantified propositions using the same engine as before.

## 3.4 $*10$ The theory of one apparent variable

**Motivation.** If $*9$ is the justification, $*10$ is the *working predicate calculus*: the laws of the single quantifier that one actually uses. A modern reader may take $*10$'s two governing rules as basic and treat $*9$ as their underpinning. Everything characteristic of quantificational reasoning—instantiation, generalization, and the distribution of quantifiers over the connectives—is established here.

**The two rules.** The pillars are universal instantiation and universal generalization,

$*\mathbf{10.1}\quad (x).\,\phi x \supset \phi y,$

$*\mathbf{10.11}\quad \text{from } \vdash \phi y \text{ infer } \vdash (x).\,\phi x,$

the latter with the usual proviso that $y$ is free for $x$ and not otherwise constrained. Existential generalization is their companion,

$*\mathbf{10.24}\quad \phi y \supset (\exists x).\,\phi x.$

**Distribution laws.** The heart of $*10$ is the interaction of the quantifier with $\supset$, $\wedge$, and a side proposition:

$*\mathbf{10.21}\quad (x).\,(p \supset \phi x) \equiv \big(p \supset (x).\,\phi x\big),$

$*\mathbf{10.22}\quad (x).\,(\phi x \wedge \psi x) \equiv (x).\,\phi x \wedge (x).\,\psi x,$

$*\mathbf{10.23}\quad (x).\,(\phi x \supset p) \equiv \big((\exists x).\,\phi x \supset p\big),$

$*\mathbf{10.28}\quad (x).\,(\phi x \supset \psi x) \supset \big((x).\,\phi x \supset (x).\,\psi x\big).$

In $*10.21$ and $*10.23$ the variable $x$ is not free in $p$; these are the laws that let a quantifier be moved past a proposition it does not concern, and that convert a universal antecedent into an existential. The universal distributes over conjunction exactly ($*10.22$) but over implication only conditionally ($*10.28$, one direction). The existential, dually, distributes over disjunction but only one way over conjunction,

$*\mathbf{10.5}\quad (\exists x).\,(\phi x \wedge \psi x) \supset (\exists x).\,\phi x \wedge (\exists x).\,\psi x,$

whose converse fails—a point $*10$ is careful to respect and a classic student trap.

## 3.5 ∗11 The theory of two apparent variables

**Motivation.** Number ∗11 generalizes to several quantifiers, and its real content is the behavior of quantifiers of *different* kinds when nested—the subtlest and most error-prone part of elementary logic. It is here that "for every $x$ there is a $y$" is carefully separated from "there is a $y$ for every $x$."

**Commuting like quantifiers.** Quantifiers of the same kind commute freely,

**∗11.2** $(x)(y).\,\phi(x,y) \;\equiv\; (y)(x).\,\phi(x,y)$,

and likewise for two existentials. Reordering a block of universals, or a block of existentials, changes nothing.

**The one-way law.** Mixed quantifiers do *not* commute; only one direction holds,

**∗11.26** $(\exists x)(y).\,\phi(x,y) \;\supset\; (y)(\exists x).\,\phi(x,y)$,

and the converse is invalid. That ∗11.26 goes one way only is the formal root of the difference between "some $x$ works for all $y$" (a single universal witness) and "each $y$ has some $x$" (a witness that may vary with $y$). *Principia Mathematica*'s scrupulous treatment of this asymmetry is one of the places where its pedantry pays off.

## 3.6 ∗12 The hierarchy of types and the axiom of reducibility

**Motivation.** Number ∗12 confronts the price of safety. To bar the paradoxes, *Principia Mathematica* has stratified functions into the ramified hierarchy: a function's *order* records the quantifiers used in defining it, and no function may be defined by quantifying over a totality that includes itself. This honors Russell's vicious-circle principle, but it is ruinous for analysis. Standard mathematical objects—the least upper bound of a bounded set of reals, the ancestral of a relation, definitions "by induction"—are *impredicative*: they quantify over collections to which the thing defined belongs, and so land in ever-higher orders, breaking the theory of real numbers. *Principia Mathematica*'s remedy is a bold assumption.

**The axiom.** The axiom of reducibility asserts that every function, of whatever order, is coextensive with a *predicative* one:

**∗12.1** $(\exists \psi).\ \phi x \equiv_x \psi! x$,

and its two-variable form ∗12.11 says the same for relations. In words: however high the order of $\phi$, there is a predicative $\psi!$ true of exactly the same things. Since mathematics cares only about *which* things a function is true of—its extension—this lets every impredicative definition be replaced, for mathematical purposes, by a predicative surrogate, and analysis is restored.

**Criticism.** No part of *Principia Mathematica* has drawn more fire, and the authors half knew it: the second edition's Introduction already calls the axiom a blemish. The objections are of two kinds. First, it looks *ad hoc*—an existence assumption introduced precisely to undo the restriction the type theory had just imposed, so that one wonders why ramify at all. Second, and worse for logicism, it looks like no law of *logic*. Ramsey (1925) argued that ramification was unnecessary to begin with, because the semantic paradoxes it also guards against are not mathematical, and that a *simple* theory of types—keeping type distinctions but dropping the order hierarchy—would need no reducibility at all; Wittgenstein objected that a genuine logical

truth cannot be, as reducibility is, merely contingently true. Hilbert and Ackermann recorded the same dissatisfaction. The verdict of history is Ramsey's: the simple theory of types, adopted by Carnap and given its canonical form by Church, became standard, and the ramified hierarchy with its axiom was set aside. Russell's own second-edition experiment (Appendix B) tried to derive mathematical induction without reducibility and is now known to contain an error. For the reader of ∗12, the practical upshot is this: reducibility is doing indispensable work in everything that follows—most immediately in the definition of identity—and it is the seam at which *Principia Mathematica*'s logicism is most visibly strained.

## 3.7 ∗13 Identity

**Motivation.** Strikingly, *Principia Mathematica* does not take identity as primitive: it *defines* it, and the definition is a formal rendering of Leibniz's principle of the identity of indiscernibles. Two objects count as identical when they share all their properties—but "all their properties" must be tamed by the type theory, and this is where the machinery of ∗12 is first cashed out.

**The definition.** Identity is defined through predicative functions,

**∗13.01** $x = y \stackrel{\text{def}}{=} (\phi).\ \phi!x \supset \phi!y,$

"$x$ is identical with $y$ if $y$ has every predicative property of $x$." The restriction to predicative $\phi!$ is essential—an unrestricted quantifier over *all* functions would be impredicative and ill-formed—but it appears to weaken the principle, since it demands agreement only on predicative properties. Here the axiom of reducibility earns its keep: because every function is coextensive with a predicative one (∗12.1), agreement on all predicative functions secures agreement on all functions whatever, and the definition recovers the full Leibnizian law. The one-directional $\supset$ suffices, since interchanging $x$ and $y$ yields the converse.

**The laws.** From the definition *Principia Mathematica* proves that identity is an equivalence relation—reflexive, symmetric, and transitive—

**∗13.15** $x = x,$

with symmetry and transitivity following at ∗13.16–∗13.17, together with the crucial law of substitutivity, that identicals may be substituted in any context ($x = y \supset (\phi x \supset \phi y)$, essentially the definition read forwards). These are the properties on which every later use of "="—and, through it, the definitions of unit class, cardinal number, and much else in ∗50 onward—will rest.

## 3.8 ∗14 Descriptions

**Motivation.** Number ∗14 is the philosophical jewel of the early *Principia Mathematica*: the formal home of Russell's theory of descriptions, first announced in "On Denoting" (1905) and widely regarded as a paradigm of analytic method. The problem is how to treat phrases of the form "*the* so-and-so." Such a phrase looks like a name, but it may name nothing—"the present King of France," "the greatest prime" —and if it were a name, sentences containing it would either be meaningless or commit us to non-existent objects. Russell's solution is to deny that descriptions are names at all: they are *incomplete symbols*, meaningful only in the context of a whole sentence, which is analyzed away in favor of quantifiers and identity.

**The contextual definition.** *Principia Mathematica* therefore defines not the description in isolation but any sentence in which one occurs. Writing $(\imath x)(\phi x)$ for "the $\phi$," the definition of "the $\phi$ is $\psi$" is

$*\mathbf{14.01}\quad \psi(\imath x)(\phi x) \stackrel{\text{def}}{=} (\exists b).\,\big[(x).\,(\phi x \equiv x = b)\big] \wedge \psi b,$

that is: there is something $b$ which is the one and only $\phi$, and $b$ is $\psi$. A description-sentence thus makes a threefold claim—*existence* (there is a $\phi$), *uniqueness* (at most one), and *predication* ($\psi$ holds of it)—none of which requires "the $\phi$" to name anything. The companion definition introduces an existence predicate,

$*\mathbf{14.02}\quad \mathrm{E!}\,(\imath x)(\phi x) \stackrel{\text{def}}{=} (\exists b).\,(x).\,(\phi x \equiv x = b),$

"the $\phi$ exists," meaning exactly that there is one and only one $\phi$.

**Scope, and the King of France.** Because a description is eliminated in context, a sentence containing both a description and a negation is *ambiguous*—the negation may fall inside or outside the analysis—and *Principia Mathematica* marks the intended reading with a scope bracket. Russell's own example makes the difference vivid. "The present King of France is not bald" can be read as

> *primary* (wide scope): there is exactly one King of France, and he is not bald
> *secondary* (narrow scope): it is not the case that there is exactly one King of France who is bald,

which, given that there is no King of France, come apart: the first is *false* (it asserts a King exists), the second *true* (it merely denies one). The scope distinction—"primary" versus "secondary occurrence"—dissolves what looks like a violation of excluded middle, and it is a permanent contribution to logic: the recognition that natural-language definite descriptions carry a scope that must be made explicit.

**Some propositions.** The number then proves the laws that make descriptions usable: that a description behaves like a genuine term *provided it exists*, so that from $\mathrm{E!}\,(\imath x)(\phi x)$ one may reason about "the $\phi$" almost as if it were a name (e.g. $*14.18$, that if the $\phi$ exists then whatever holds of everything holds of it); that any true predication of the $\phi$ guarantees its existence ($\psi(\imath x)(\phi x) \supset \mathrm{E!}\,(\imath x)(\phi x)$, in the primary sense); and the identity conditions of descriptions. The moral throughout is the one Russell prized: apparent reference to a dubious object has been *paraphrased away*, leaving only quantifiers over ordinary things.

**Significance.** The theory of descriptions became the model for a whole style of philosophy—analysis as the replacement of a misleading surface form by a logically perspicuous one, with ontological economy as the reward. Its influence runs through Quine's programme and the mid-century debates on reference; its most famous challenge, Strawson's "On Referring" (1950), argued that Russell had mistaken a presupposition of reference for part of what is asserted. Whatever one makes of that dispute, $*14$ remains the point where *Principia Mathematica*'s formal machinery visibly earns its philosophical keep.

## 3.9 Closing notes

**The $*8$ alternative.** The second edition, persuaded by Wittgenstein that genuine generality and the stroke should be taken more seriously, replaced the main-text $*9$ with a new $*8$ in Appendix A, handling propositions with apparent variables differently and permitting real

variables to be understood more uniformly. The abridged *to* ∗56 prints both, and a careful reader should know that the "official" quantification theory of the first edition (∗9) and the second edition's revision (∗8) are not identical; the working results of ∗10–∗11 are common ground.

**The through-line.** These six numbers are held together by a single strain: the paradoxes force the ramified hierarchy; the hierarchy breaks mathematics; reducibility repairs it at the cost of logical purity; and identity and descriptions are then built on that repaired foundation. It is the stretch of *Principia Mathematica* where the grandeur and the fragility of the logicist project are both most apparent.

**Ahead.** Installment III turns to classes and relations (∗20–∗43): the "no-class" theory that treats classes, like descriptions, as incomplete symbols; the calculus of classes and of relations; and the groundwork for the cardinal arithmetic of ∗50–∗56 with which the abridged volume closes.

**Sources.** Formal statements above follow *Principia Mathematica* (2nd ed., 1925–27) and the abridged *to* ∗56; for the definitions of identity (∗13.01), reducibility (∗12.1), and descriptions (∗14.01–∗14.02), and for the predicative-function ("shriek") notation, I have followed B. Linsky's "The Notation in *Principia Mathematica*" and A. D. Irvine's "*Principia Mathematica*," both in the *Stanford Encyclopedia of Philosophy*, checked against the Gutenberg text of Volume I. The critical discussion of reducibility draws on Ramsey (1925), Church (1976), and the standard histories.

# Chapter 4

# Classes and Relations (∗20–∗43)

*This installment covers* Principia Mathematica*'s theory of classes and relations—the numbers in which the book, having built a logic of quantified propositional functions, constructs the extensional apparatus on which its arithmetic will rest. Four blocks are treated: the "no-class" theory that renders classes and relations as incomplete symbols (∗20–∗21); the Boolean calculus of classes and relations (∗22–∗25); the calculus of relations proper—descriptive functions, converses, domains and fields, relative products, restrictions, and images (∗30–∗38); and the general products and sums of classes of classes (∗40–∗43). This is the most technical stretch of the abridged volume, and the commentary groups the numbers by function, treating the pivotal definitions in detail and the surrounding calculus more briskly.*

## 4.1 Introduction: the extensional turn

With ∗14 Russell had shown how to speak of "the so-and-so" without positing a mysterious object to be its meaning. The numbers now before us apply the same strategy twice more, to *classes* and to *relations*. A class such as "the class of prime numbers" looks like a name for a thing—a collection—and classes, taken as things, had been the source of the paradox. *Principia Mathematica*'s response is its celebrated *no-class theory*: class expressions are incomplete symbols, defined only in context and analyzed away in favor of the propositional functions that determine them. There are, strictly, no classes in *Principia Mathematica*'s ontology; there are only functions and a systematic way of talking *as if* there were classes.

Two features make this workable. The first is *extensionality*: for the purposes of the class calculus, functions true of exactly the same things are interchangeable, so that a class is fixed by its members and nothing else. The second is, once again, the axiom of reducibility, which guarantees that the predicative function needed to represent a class is always available. On this basis *Principia Mathematica* recovers the whole Boolean algebra of classes (∗22), and—because relations are treated in exactly parallel fashion, as classes of ordered couples—the algebra of relations as well (∗23).

The larger purpose is arithmetic. The later volumes will define a cardinal number as a class of equinumerous classes, and an ordinal as a class of similar well-ordered relations; series, limits, and continuity will all be relational. So the relational vocabulary developed here—the *domain* and *field* of a relation, its *converse*, the *relative product* that composes two relations, the *image* of a class under a relation—is not idle machinery but the toolkit in which numbers and order are about to be built. This installment is, in effect, the workshop.

## 4.2 Notation: classes, relations, and the relational toolkit

**Classes.** $\hat{z}(\phi z)$ denotes "the class of $z$ such that $\phi z$," and membership is written with the Greek epsilon (for *ἐστί*, "is a"), which we render by the modern $\in$: $x \in \hat{z}(\phi z)$ means $x$ satisfies $\phi$. The class relations are inclusion $\alpha \subset \beta$, and the Boolean operations product $\alpha \cap \beta$, sum $\alpha \cup \beta$, and complement $-\alpha$ (within a type). $V$ is the universal class, $\Lambda$ the null class, and $\exists!\,\alpha$ says $\alpha$ has a member.

**Relations.** $\hat{x}\hat{y}\,\phi(x,y)$ denotes the relation in extension holding between $x$ and $y$ when $\phi(x,y)$. The relational operations we shall meet are the *descriptive function* $R$`` `y `` ("the $R$ of $y$"), the *converse* $\breve{R}$, the *domain* $D$`` `R ``, the *converse domain* $\mathfrak{C}$`` `R ``, the *field* $C$`` `R ``, the *relative product* $R \mid S$, and the *image* $R$`` ``\beta `` of a class $\beta$ under $R$. The raised comma in $R$`` `y ``, $D$`` `R ``, $C$`` `R `` is *Principia Mathematica*'s notation for applying a function to an argument.

## 4.3 ∗20–∗21 The no-class theory of classes and relations

**Motivation.** These two numbers install the extensional framework. Their task is to define class expressions and relation expressions so that they can be used freely—in membership statements, in the Boolean calculus, as arguments to further functions—while committing the system to no classes or relations as entities. It is the theory of descriptions again, generalized from "the $\phi$" to "the class of $\phi$'s."

**Classes as incomplete symbols.** A sentence containing a class term is defined by eliminating the term in favor of the function determining it. Schematically, to assert something $f$ of the class $\hat{z}(\psi z)$ is to assert that some predicative function coextensive with $\psi$ satisfies $f$; this is *Principia Mathematica*'s ∗20.01, and it leans on reducibility to supply the predicative representative. The one consequence a reader must carry forward is the membership law,

**∗20.02** $\quad x \in \hat{z}(\phi z) \;\equiv\; \phi x,$

that to be a member of the class of $\phi$'s is just to be a $\phi$. From the contextual definition *Principia Mathematica* derives *extensionality*, that coextensive functions determine the same class,

**∗20.15** $\quad \big[(x).\,\phi x \equiv \psi x\big] \;\supset\; \hat{z}(\phi z) = \hat{z}(\psi z),$

so that a class is individuated by its members alone. The philosophical point is carried in the name: this is the *no-class* theory, on which classes are *logical fictions*, useful abbreviations with no independent being.

**Relations in extension.** Number ∗21 repeats the construction for dyadic relations, treating a relation in extension as fixed by the couples it holds between—the relational analogue of a class being fixed by its members. Where ∗20 writes $\hat{z}(\phi z)$, ∗21 writes $\hat{x}\hat{y}\,\phi(x,y)$, with a parallel contextual definition and a parallel extensionality: relations holding between exactly the same pairs are identified. Everything that follows applies the two theories in tandem.

## 4.4 ∗22–∗25 The Boolean calculus

**Motivation.** With classes available, *Principia Mathematica* develops their algebra—and in doing so recovers, inside its logicist system, the algebra of logic that Boole, Peirce, and Schröder

had studied. Numbers ∗22–∗25 are where *Principia Mathematica*'s notation for "⊂," "∩," "∪" is fixed and their laws proved.

**The operations and their laws.** Inclusion and the Boolean operations are defined by

**∗22.01** $\alpha \subset \beta \stackrel{\text{def}}{=} (x).\,(x \in \alpha \supset x \in \beta)$,

**∗22.02–04** $\alpha \cap \beta \stackrel{\text{def}}{=} \hat{x}(x \in \alpha \wedge x \in \beta), \quad \alpha \cup \beta \stackrel{\text{def}}{=} \hat{x}(x \in \alpha \vee x \in \beta), \quad -\alpha \stackrel{\text{def}}{=} \hat{x}(x \notin \alpha)$,

and from them *Principia Mathematica* proves the full complement of Boolean laws—commutativity, associativity, the distributive laws, and the De Morgan laws for classes—each a direct transcription of the corresponding propositional law of ∗3–∗4 under the membership equivalence ∗20.02. The calculus of classes is, in this precise sense, the propositional calculus wearing extensional clothing.

**Universal and null classes.** Number ∗24 introduces the two extreme classes and the existence predicate,

**∗24.01–03** $V \stackrel{\text{def}}{=} \hat{x}(x = x), \qquad \Lambda \stackrel{\text{def}}{=} \hat{x}(x \neq x), \qquad \exists!\,\alpha \stackrel{\text{def}}{=} (\exists x).\,x \in \alpha$,

so that $V$ is the class of everything (of a given type), $\Lambda$ the class of nothing, and $\exists!\,\alpha$ the assertion that $\alpha$ is non-empty. The expected laws follow: $\Lambda \subset \alpha \subset V$, $\alpha \cup -\alpha = V$, $\alpha \cap -\alpha = \Lambda$. Number ∗25 supplies the strict analogues for relations—the universal relation, the null relation, and the existence of relations—closing the parallel between the two calculi.

## 4.5 ∗30–∗38 The calculus of relations

**Motivation.** Here *Principia Mathematica* builds the working theory of relations that the rest of the enterprise runs on. A relation, for *Principia Mathematica*, is the fundamental order-bearing notion—succession, greater-than, membership itself—and this block gives the operations for handling relations as objects of calculation: turning a relation into a function, reversing it, extracting the things it relates, composing two relations, and applying a relation to a whole class.

**Descriptive functions.** The pivotal definition—historically the first modern account of a *function* as a special kind of relation—is that of the descriptive function,

**∗30.01** $R`y \stackrel{\text{def}}{=} (\imath x)(x\,R\,y)$,

"the $R$ of $y$": the unique $x$ bearing $R$ to $y$, if there is one. This ties the theory of relations directly to the theory of descriptions of ∗14: "the father of $y$," "the successor of $n$," "the limit of $s$" are all descriptive functions, defined and well-behaved exactly when the relevant $x$ exists and is unique. It is the mechanism by which *Principia Mathematica* recovers ordinary functional notation inside a logic whose primitive is the relation.

**Converse, domain, field.** Three further operations extract structure from a relation. The converse reverses it,

**∗31.01** $x\,\breve{R}\,y \equiv y\,R\,x$,

and the domain, converse domain, and field collect the terms it relates,

**∗33.01–03** $D`R \stackrel{\text{def}}{=} \hat{x}\{(\exists y).\,x\,R\,y\}, \quad \mathrm{C}`R \stackrel{\text{def}}{=} \hat{y}\{(\exists x).\,x\,R\,y\}, \quad C`R \stackrel{\text{def}}{=} D`R \cup \mathrm{C}`R$,

the domain being the class of things that bear $R$ to something, the converse domain the

class of things something bears $R$ to, and the field their union. (The intervening $*32$ isolates, for a fixed term, the classes of its referents and relata.) These give the vocabulary in which "the numbers less than $n$" or "the ancestors of $x$" are expressed.

**Relative product.** Composition of relations is the relative product,

$*\mathbf{34.01}\quad x\,(R \mid S)\,z \;\equiv\; (\exists y).\,x\,R\,y \wedge y\,S\,z,$

"$x$ is an $R$ of an $S$ of $z$"—as "maternal grandmother" is "mother" relative-multiplied by "mother." Its powers $R^2 = R \mid R$, and so on, will be the raw material of the ancestral relation and hence of mathematical induction in the numbers beyond $*56$.

**Restrictions and images.** The remaining numbers of the block confine and apply relations. A relation may be limited to a class in its domain or converse domain ($*35$–$*36$); and, most usefully, a relation carries a whole class to its image,

$*\mathbf{37.01}\quad R\text{`}\text{`}\beta \stackrel{\text{def}}{=} \hat{x}\{(\exists y).\,y \in \beta \wedge x\,R\,y\},$

the *plural descriptive function* $R\text{`}\text{`}\beta$: the class of all $R$'s of members of $\beta$—"the parents of the Europeans," "the successors of the evens." Restriction and image are exactly what let one speak of a relation's action on a set, the relational counterpart of the image of a set under a map. (Number $*38$ adds bookkeeping for treating dyadic operations, like $\alpha + \beta$, relationally.)

## 4.6 $*40$–$*43$ Products and sums of classes of classes

**Motivation.** The final block generalizes intersection and union from two classes to an entire *class of classes*—the operation of intersecting, or uniting, all the members of a family at once. This is indispensable for what follows: defining a cardinal sum requires uniting a family of disjoint classes, and the theory of series will repeatedly take the product or sum of a whole collection.

**General product and sum.** For a class of classes $\kappa$,

$*\mathbf{40.01}$–$\mathbf{02}\quad p\text{`}\kappa \stackrel{\text{def}}{=} \hat{x}\{(\alpha).\,\alpha \in \kappa \supset x \in \alpha\}, \qquad s\text{`}\kappa \stackrel{\text{def}}{=} \hat{x}\{(\exists\alpha).\,\alpha \in \kappa \wedge x \in \alpha\},$

the product $p\text{`}\kappa$ being the class of things belonging to *every* member of $\kappa$, and the sum $s\text{`}\kappa$ the class of things belonging to *some* member. These are the arbitrary intersection $\bigcap\kappa$ and union $\bigcup\kappa$ of modern set theory. Number $*41$ supplies the relational analogues (the product and sum of a class of relations), and $*42$–$*43$ collect miscellaneous results, including the interaction of the relative product with these general operations—technical lemmas that the later construction of arithmetic will draw on.

## 4.7 Closing notes

**What has been built.** Across $*20$–$*43$ *Principia Mathematica* has erected a complete extensional logic: classes and relations introduced as incomplete symbols resting on reducibility, a Boolean algebra of each, and a rich calculus of relations with functions, converses, fields, composition, and images. None of it enlarges the ontology—every class and relation term is in principle eliminable—yet it furnishes exactly the notions in which number and order can be defined. The extensional turn is what converts the logic of $*1$–$*14$ into a mathematics.

**Ahead.** The abridged volume closes with ∗50–∗56: the diversity relation, unit classes, the cardinal couple, and the first steps of cardinal arithmetic—where the apparatus assembled here is finally turned to defining numbers themselves. The concluding installment will treat those numbers, after which the four parts, together with the technical paper on the Logic Theorist, can be assembled into a single volume.

**Sources.** Formal statements follow *Principia Mathematica* (2nd ed.) and the abridged *to* ∗56, with notation and numbering checked against the Wikipedia *Glossary of Principia Mathematica*, B. Linsky's "The Notation in *Principia Mathematica*" and A. D. Irvine's "*Principia Mathematica*" (both *Stanford Encyclopedia of Philosophy*), and the Gutenberg text of Volume I; the reading of the no-class theory draws on the standard scholarship of Landini and Linsky.

# Chapter 5

# Unit Classes and Couples: the First Cardinals (∗50–∗56)

*This concluding installment reaches the destination toward which the whole abridged volume has been building: the definition of* number. *The numbers* ∗50–∗56 *form* Principia Mathematica*'s "Prolegomena to Cardinal Arithmetic," and in them identity and diversity are recast as relations (*∗50*), the unit class is defined (*∗51*), and the first cardinals are constructed—the number* 1 *as the class of all unit classes (*∗52*), and* 0 *and* 2 *by way of cardinal couples (*∗54*)—culminating in the celebrated* ∗54.43*, the precursor of "*$1 + 1 = 2$*." Ordered pairs then appear as* ordinal *couples (*∗55*), and the first relation-number, the ordinal* $2_r$*, distinguishes ordinal from cardinal (*∗56*). Here the logicist reduction of arithmetic to logic makes its first concrete payment.*

## 5.1 Introduction: the threshold of arithmetic

The abridged *Principia Mathematica* ends where arithmetic begins. Everything to this point—the propositional calculus, quantification and types, identity and descriptions, the extensional theory of classes and relations—has been apparatus. Numbers ∗50–∗56 put that apparatus to its first mathematical use, defining the smallest cardinals and the ordered pair, and reaching, in ∗54.43, the exact threshold of the claim that $1 + 1 = 2$.

The governing idea is the Frege–Russell conception of number, and it is worth stating plainly because it can seem strange. A cardinal number is not, for *Principia Mathematica*, a primitive object; it is a *property of classes*—the property two classes share when their members can be paired off one-to-one—and *Principia Mathematica* reifies that property in the only material it allows itself, as the *class of all classes* having it. The number 1 *is* the class of all one-membered classes; the number 2 *is* the class of all two-membered classes; a number counts by being the collection of all the collections of that size. This is logicism cashed out: numbers turn out to be logical objects, definable with no vocabulary beyond that of classes and relations.

*Principia Mathematica* approaches this in miniature here, defining 0, 1, and 2 directly (the general definition of a cardinal, as the class of all classes similar to a given one, awaits ∗100 in Volume II). By ∗56 it has just enough—the numbers up to two, and the ordered pair—to make the cardinal/ordinal distinction and to stand at the door of arithmetic proper. That is why the abridged volume stops here: it is a natural resting point, the moment the machine first produces a number.

## 5.2 ∗50 Identity and diversity as relations

**Motivation.** Identity was defined in ∗13 as a propositional function. To use it in the calculus of relations—as a thing with a domain, a converse, a relative product—it must be recast as a *relation in extension*. Number ∗50 does this, and introduces its complement, diversity.

**∗50.01** $I \stackrel{\text{def}}{=} \hat{x}\hat{y}\,(x = y)$,

**∗50.02** $J \stackrel{\text{def}}{=} -I$,

so that $I$ is the identity-relation (each thing bears $I$ to itself and nothing else) and $J$, its relational complement, is diversity, the relation each thing bears to everything other than itself. These will recur constantly: the unit class of $x$, next, is essentially the class of things identical with $x$, and "two" will be built on things being *diverse*.

## 5.3 ∗51 Unit classes

**Motivation.** Before there can be a number one, there must be the classes it counts: the *unit classes*, or singletons. Number ∗51 defines the singleton and—crucially—insists on the distinction between an object and its unit class.

**∗51.01** $\iota`x \stackrel{\text{def}}{=} \hat{y}\,(y = x)$,

the class $\iota`x$ whose sole member is $x$. The single most important point in the number, and a persistent source of confusion it is careful to prevent, is that $x$ and $\iota`x$ are *not the same*: $x$ is a member, $\iota`x$ is a class, and they live at different type levels. (Confounding the two is precisely the kind of type-error the hierarchy of Installment II exists to forbid.) Whitehead and Russell even remark that this distinction, though it may look like pedantry, is one whose neglect breeds contradiction. The unit class is a genuine act of construction: it lifts an object into the world of classes, where it can be counted.

## 5.4 ∗52 The cardinal number 1

**Motivation.** Here *Principia Mathematica* defines its first number. The definition is the paradigm case of the Frege–Russell idea, and everything about the treatment of number in the book is visible already in it.

**∗52.01** $1 \stackrel{\text{def}}{=} \hat{\alpha}\,\{(\exists x).\,\alpha = \iota`x\}$,

the number 1 is the class of all $\alpha$ that are the unit class of something— that is, the class of all one-membered classes. Note what has and has not been done. *Principia Mathematica* has *not* produced an object called "one" standing behind the numeral; it has produced the extension of the property *being a singleton*. To say "$\alpha$ has one member" is just to say $\alpha \in 1$. Counting has been reduced to membership, and the number to a class—a reduction of arithmetic to logic in its first and cleanest instance. The number is then shown to be non-empty and to obey the laws one expects of "one"; these are the propositions ∗52 goes on to prove.

## 5.5 ∗53 Miscellaneous propositions involving unit classes

**Motivation.** A short, preparatory number. It collects the facts about unit classes—their interaction with $V$, $\Lambda$, union, and intersection, and with the existence predicate—that the

construction of couples in ∗54 will need. It is to ∗52 what ∗5 was to ∗4: bookkeeping in the service of what follows, and a reader may pass through it quickly.

## 5.6 ∗54 Cardinal couples

**Motivation.** Number ∗54 builds the two-membered classes and, with them, the numbers 0 and 2. It is the home of the volume's most famous proposition.

**Zero and two.** The extreme and the next cardinal are defined in the same style as 1,

**∗54.01** $0 \stackrel{\text{def}}{=} \iota`\Lambda,$

**∗54.02** $2 \stackrel{\text{def}}{=} \hat{\alpha}\{(\exists x)(\exists y).\, x \neq y \wedge \alpha = \iota`x \cup \iota`y\}.$

Zero is the class whose only member is the empty class—the one and only class with no members—so that 0 is a unit class, though a number; and 2 is the class of all classes consisting of two *distinct* things, a union of two unit classes with $x \neq y$. The clause $x \neq y$ is where diversity (∗50) earns its place: without it, $\iota`x \cup \iota`y$ might collapse to a singleton.

**The famous proposition.** *Principia Mathematica* then proves

**∗54.43** $\alpha, \beta \in 1 \;\supset\; \big(\alpha \cap \beta = \Lambda \;\equiv\; \alpha \cup \beta \in 2\big),$

in words: if $\alpha$ and $\beta$ are each unit classes, then they are disjoint if and only if their union is a couple. To this proposition Whitehead and Russell append the note that, once arithmetical addition has been defined, it will follow that $1 + 1 = 2$. The connection is exact. Cardinal addition (defined only in Volume II, at ∗110) takes two disjoint representative classes, one from each summand, and asks which cardinal their union falls under; to compute $1 + 1$ one takes two disjoint unit classes and unites them, and ∗54.43 guarantees the result is a couple—a member of 2—whichever representatives one chose. The identity $1 + 1 = 2$ is thus latent here, and discharged in Volume II at ∗110.643, beneath which stands the book's driest and most quoted line: "The above proposition is occasionally useful.

**The legend.** This is the origin of the popular image of *Principia Mathematica* as the book that "takes hundreds of pages to prove $1 + 1 = 2$." The image is half right. The result does arrive late, because *Principia Mathematica* refuses to assume any of arithmetic and insists on defining number, addition, and the couple from logic alone; but the length is the whole point—it is the measure of how much logic a single arithmetical truth turns out to contain.

## 5.7 ∗55 Ordinal couples

**Motivation.** The couple of ∗54 is *unordered*: $\iota`x \cup \iota`y$ is the same class as $\iota`y \cup \iota`x$. But mathematics constantly needs *ordered* pairs—first and second, antecedent and consequent, argument and value. Number ∗55 supplies the ordered pair, and does so in a way characteristic of *Principia Mathematica*: not as a class, but as a *relation*.

**∗55.01** $x \downarrow y \stackrel{\text{def}}{=} \hat{z}\hat{w}\,(z = x \wedge w = y),$

the ordinal couple $x \downarrow y$ is the relation holding *only* between $x$ (as referent) and $y$ (as relatum), and in that order. Because $x \downarrow y$ and $y \downarrow x$ are different relations, order is captured. It is worth pausing on this: the ordered pair, which modern set theory builds by Kuratowski's device $\{\{x\}, \{x, y\}\}$, appears in *Principia Mathematica* two decades earlier as a primitive-looking

relation. *Principia Mathematica* makes the ordered pair fundamental and the relation primary, where later set theory makes the class primary and defines the pair within it—a small but telling difference in what each system treats as bedrock.

## 5.8 ∗56 The ordinal number $2_r$

**Motivation.** Finally, the volume distinguishes the ordinal from the cardinal at its first opportunity. The cardinal 2 counts unordered pairs; the ordinal $2_r$ concerns the *ordered* ones, and is defined as their class.

**∗56.01** $2_r \stackrel{\text{def}}{=} \hat{R}\{(\exists x)(\exists y).\, x \neq y \wedge R = x \downarrow y\}$,

the class of all ordinal couples—all relations that are a single ordered pair of distinct terms. The subscript $r$ marks a *relation-number*: where a cardinal is a class of similar classes, an ordinal (relation-number) is a class of *ordinally similar relations*. That cardinal 2 and ordinal $2_r$ are different objects—one a class of classes, the other a class of relations—is not a defect but the first appearance of a distinction that pervades mathematics: two things versus two things *in an order*. *Principia Mathematica* reaches exactly this distinction, and no further, before the abridged volume closes.

## 5.9 Closing notes

**What the volume has accomplished.** From five propositional axioms and a single rule of inference, *to* ∗56 has traveled to the definition of the numbers 0, 1, and 2, the ordered pair, and the cardinal/ordinal distinction— the entire journey conducted in logical vocabulary, positing no numbers, no classes, and no relations as primitive entities. Whether the journey *vindicates* logicism is, as we have seen, contestable: it leans at crucial points on the axiom of reducibility and, beyond this volume, on the axiom of infinity, neither of which is obviously a truth of logic. But as a demonstration that arithmetic *can* be spun from logical thread, the construction that ends here is without precedent.

**Beyond ∗56.** Volume II opens the cardinal arithmetic proper: the general definition of a cardinal number (∗100), the operations of addition, multiplication, and exponentiation (∗110–∗117)—where $1 + 1 = 2$ is at last discharged—and the theory of finite and infinite (∗118–∗126), which is where the axiom of infinity must be invoked. Volume III reaches the ordinals and the real numbers. The abridged reader, having watched number *begin*, is left at the most inviting possible cliffhanger.

**On assembly.** With this installment the commentary spans ∗1–∗56 in four parts: the theory of deduction (∗1–∗5); quantification, types, identity, and descriptions (∗9–∗14); classes and relations (∗20–∗43); and unit classes and couples (∗50–∗56). These, together with the technical retrospective on the Logic Theorist—the program that first read this book by machine—can now be gathered into a single volume, with unified front matter, notation, and pagination.

**Sources.** Formal statements follow *Principia Mathematica* (2nd ed.) and the abridged *to* ∗56; the statement of ∗54.43 and its note were checked against the University of Michigan scan of Volume I (p. 379), and section titles against the volume's own table of contents. Notation and numbering were checked against the Wikipedia *Glossary of Principia Mathematica* and B. Linsky's "The Notation in *Principia Mathematica*" (*Stanford Encyclopedia of Philosophy*);

the account of the Frege–Russell definition of number and of the $1 + 1 = 2$ result follows the standard literature and A. D. Irvine's *SEP* entry.

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